

W4 OS Lab

1WA24CS241

MULTI QUEUE Scheduling :

```
#include <stdio.h>

int main() {
    printf("Name: Stawan H. Ruiwale\nUSN:1WA24CS241\n\n");

    int n, curr_time = 0, i, j, ns = 0, nu = 0, type[10];
    int sysPID[10], sysAT[10], sysBT[10], sysCT[10], sysTAT[10], sysWT[10];
    int usrPID[10], usrAT[10], usrBT[10], usrCT[10], usrTAT[10], usrWT[10];
    double tat_sum = 0, wt_sum = 0;

    printf("Enter no. of processes: ");
    scanf("%d", &n);

    // input
    for (i = 0; i < n; i++) {

        // process type
        printf("Enter type for process[%d] (0=system, 1=user): ", i + 1);
        scanf("%d", &type[i]);

        // at and bt of the process
        printf("Enter AT and BT for process[%d]: ", i + 1);

        // storing the info in relevant type
        if (type[i] == 0) {
            scanf("%d %d", &sysAT[ns], &sysBT[ns]);
            sysPID[ns] = i + 1;
            ns++;
        } else {
            scanf("%d %d", &usrAT[nu], &usrBT[nu]);
            usrPID[nu] = i + 1;
            nu++;
        }
    }
}
```

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}

// system queue (FCFS)
for (i = 0; i < ns; i++) {

    if (curr_time < sysAT[i])
        curr_time = sysAT[i];

    curr_time += sysBT[i];

    // CT
    sysCT[i] = curr_time;

    // TAT
    sysTAT[i] = sysCT[i] - sysAT[i];
    tat_sum += sysTAT[i];

    // WT
    sysWT[i] = sysTAT[i] - sysBT[i];
    wt_sum += sysWT[i];
}

// user queue (FCFS)
for (i = 0; i < nu; i++) {

    if (curr_time < usrAT[i])
        curr_time = usrAT[i];

    curr_time += usrBT[i];

    // CT
    usrCT[i] = curr_time;

    // TAT
    usrTAT[i] = usrCT[i] - usrAT[i];
    tat_sum += usrTAT[i];

    // WT
    usrWT[i] = usrTAT[i] - usrBT[i];
    wt_sum += usrWT[i];
}

```

```

// header row
printf("\nPID\tAT\tBT\tType\tCT\tTAT\tWT\n\n");

for (j = 1; j <= n; j++) {

    // search system queue
    for (i = 0; i < ns; i++) {
        if (sysPID[i] == j) {
            printf("P%d\t%d\t%d\tSystem\t%d\t%d\t%d\n", sysPID[i], sysAT[i],
                sysBT[i], sysCT[i], sysTAT[i], sysWT[i]);
        }
    }

    // search user queue
    for (i = 0; i < nu; i++) {
        if (usrPID[i] == j) {
            printf("P%d\t%d\t%d\tUser\t%d\t%d\t%d\n", usrPID[i], usrAT[i], usrBT[i],
                usrCT[i], usrTAT[i], usrWT[i]);
        }
    }
}

printf("\nAverage TAT: %.2f\n", tat_sum / n);
printf("Average WT: %.2f\n", wt_sum / n);

return 0;
}

```

Output :

Name: Stawan H. Ruiwale
USN:1WA24CS241

Enter no. of processes: 4
Enter type for process[1] (0=system, 1=user): 0
Enter AT and BT for process[1]: 0 4
Enter type for process[2] (0=system, 1=user): 0
Enter AT and BT for process[2]: 0 3
Enter type for process[3] (0=system, 1=user): 1
Enter AT and BT for process[3]: 0 8
Enter type for process[4] (0=system, 1=user): 0
Enter AT and BT for process[4]: 10 5

PID	AT	BT	Type	CT	TAT	WT
P1	0	4	System	4	4	0
P2	0	3	System	7	7	4
P3	0	8	User	23	23	15
P4	10	5	System	15	5	0

Average TAT: 9.75

Average WT: 4.75

stavvv@Stavs-MacBook-Air Desktop %